

20260720 Cluster access, project Management, gLike on Cluster, NH meeting

Cluster access

- I set up the USC VPN on both my laptop and home machine
- I can confirm access to `discovery.usc.edu` from on-campus wifi (as of last Friday)
- To access from off-campus, I need to use a windows powershell and directly connect to the ip of `discovery2.usc.edu`, which is pretty annoying but it still works I guess.
- I'm getting a new laptop soon. @jonmah make sure to redo this.

Project management

[image]I can confirm access to CARC project management and have opened a project called gLike Optimization. I assigned @chia657 as a project manager.

- Field of science: Genetics and Nucleic Acids
- Project Security: USC Clusters
- Project Status: Active
- Created: July 20, 2026
- Billing Admin Dept: Center for Genetic Epidemiology
- Billing Admin Name: Kimberly Louie
- Billing Admin Email: Kimberly.Louie@med.usc.edu
- Billing Admin Phone: (323) 442-3930
- Billing Cost Center: None

- Billing PPGG: None
- FBS account worktag: None

I have a single allocation under resource: `Discovery` (Cluster partition) with the slurm account `mahj_1946`. I requested "1 million hours" (free, which was approved), for an annual cost of \$0 and 15TB of storage.

gLike on Cluster (WIP)

I am currently installing / setting up a workspace on the USC cluster
(**making sure not to compute on the login node**)

Native Hawaiian (NH) meeting

- Charleston's daughter is home-free roaming around for this week, so meeting schedule is wack
- Weird ai robot is following me with a dead link --@jonmah look into this?
 - Attached to usc acc?
- Xinran update:
 - Citations update for trang
 - BMI GWAS, has the result changed with the inclusion of additional individuals?
 - Loading data might take a while, issues with compute power availability
 - Sycamore / Relate issue
 - There are no individual names in common between the ARG and phenotype file
 - Try: double-id issue?
 - ID_1 is ID_2, and phenotype file is also changed to be the same
 - Some troubleshooting
 - Proceed with newest version of sycamore
 - issues seem to be with lab id matching
 - System is horrible with lab ids (primary lab id system vs lab

id system)

- Primary lab id system is one person per id
- Lab id system is one sample per id
- Much of the older genetic id is in the form of lab id (one per sample), but newer data is based on primary lab id (one sample per person)
- Format: MXXXXXX where X is some number
- Is this a primary lab id vs. lab id issue? No overlap vs. incomplete overlap of id?
 - Try: verify if id system systems are the same
 - Try: Does the order matter? It does in a Relate tree (rearrange the phenotype file)
 - Question from Bryan: Does the script check both IDs?
- Bryan update:
 - Indexing error due to SNP IDs
 - Multiallelic variants
 - Same command as with NH and samoan data
 - Tested on chunk without multiallelic variants
 - Re-ran whole chromosome with `-m none` argument
 - Condensed Kimberli's pipeline into one script
 - Ran for everything (runs in one day due to condensed script)
 - Some arrays didn't start, but they did work on different partitions
 - `conti` partition --> `chiang` partition
 - `mega/gda/p13/p77/p99` data
 - Somoan data
 - `addPol` data
 - Kimberli is also running `addPol` data so to compare scores when done
 - 14 hits with 3 cohoroats (not with `addPol`)
 - 5 hits with 4 cohorts (including `addPol`)
 - `addPol` is a code for "additional Polynesians"
 - Look into chromosome 2?
 - @jonmah look into Bryan Dinh's PLoS Genetics paper
 - Waiting in some element on `AllofUs` data to download certain summary statistics
- Eunice update:
 - Working on comments for poster for some sort of

conference

- Polygenic scores for metabolic traits (e.g., BMI) explain limited phenotypical variance
 - inaccurate when applied to non-European populations
 - dietary patterns are established metabolic risk factors
 - how does cross-ancestry PGS + dietary information provide insight into BMI for five different (multi-ethnic) ancestry groups
- Primary lab id vs. lab id strikes again (for the poster things in lab id is fine)
- Ji update:
 - paper updates
- Eaaswar
 - MEC-NH Parents (parenthood)
 - Informative individuals by group for a variety of metrics, e.g.,
 - Individuals with at least one Filipino parent
 - Individuals with exactly one Filipino parent
 - Individuals with both Filipino parents
 - ^ ^ ^ ^ ^ same for EastAsian parenthood
 - Ethnicity flags for parent hood
 - Doublecheck to make sure that everyone reports parental ethnicity
 - Tracts 2.0
 - Demographic modeling framework using YAML-based model and driver files
 - Sex based migration
 - Sex-specific recombination
 - Joint modeling of autosomes
 - Results are comparison of a variety of models
 - Founding time
 - Founding POL, Founding FIL
 - NFE pulse time / proportion
 - NFE implies Filipino heritage
 - EAS pulse time / proportion
 - EAS implies East Asian
 - In prior analysis, all three algorithms ultimately infer a large amount of Filipino ancestry

Bryan deliverable

- Bryan was very helpful in providing instructions for how to access the NH data at `/project2/chia657_28/datasets/private/MEC/WGS/CCDG_WGS/pass_subset_10721_geno02_dropSingletons`.
 - However, I appear to have run into some sort of issue where I can navigate to the file, but do not have read or write permissions for this specific directory.
 - [image]Image: the directory name is a different color and cannot be accessed
 - based on the first character for the filetype string, it's a symbolic link to a different directory
 - [image]When I follow the symbolic link to a different directory, I find that I also have my read access permission denied.
 - @jonmah reach out to Bryan Dinh and/or Charleston

TODO

- Deliverable for @jonmah: by **end of august**, aiming to have first version of Eaaswar + Ji paper
 - fused multivariate VCF format figured out, tentatively by 3rd week of august
 - after first draft , include additional 500 people
- Coordinate with Xinran and Bryan re NH data (**Also sometime around end of August**)
 - Do I actually have access to: `/project2/chia657_28/datasets/private/MEC/WGS/CCDG_WGS/pass_subset_10721_geno02_dropSingletons?`
- Follow-up with Nate (Nstitzel)
- Follow-up with Christian Fuchsberger
- USCard should be available and active now, pick-up tomorrow